Machine Learning Assisted Real-Time Plasma Diagnosis: Domain-Knowledge-Driven Feature Engineering Enable H_2O_2 Yield Prediction

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Introduction

Nanosecond pulsed CO2 bubble discharge demonstrates novel pathway that utilizes plasma technology to convert CO2 into useful chemical products, such as H2 and H2O2[1].

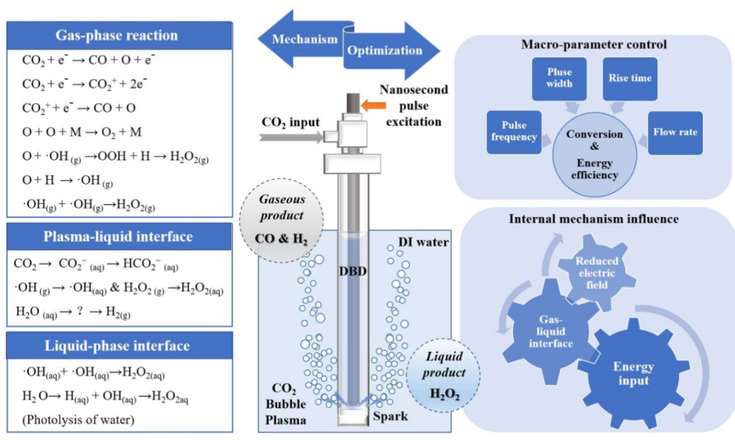

Figure 1: Nanosecond Pulsed CO_2 Bubble Discharge Setting[1]

Optical Emission Spectroscopy (OES) is a cost-effective instrument that enables non-intrusive, real-time measurement into the status of plasma.

In this research, **OES spectrum, combined with reaction discharge parameters**, are engineered to build informative features that allow Machine Learning models (i.e. linear regression, artificial neural network) to recognize and **predict H2O2 yield**. Data used in this research is collected from Gao et al.[1]

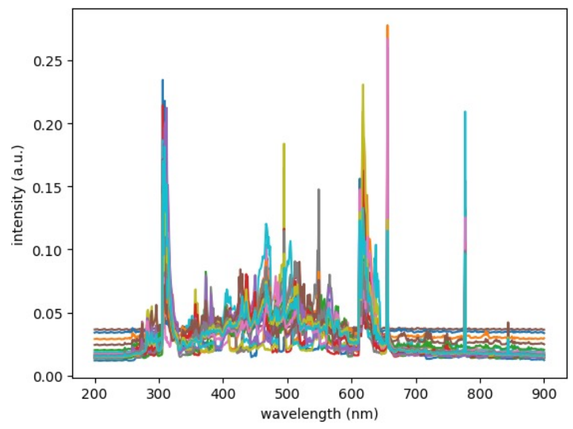

Figure 2: 20 Optical Emission Spectrum of 701 Samples

Methodology

The project is conducted in **4 iterative phases**, each building on the previous one to enhance model performance and interpretability.

Phase 1: Baseline modeling with PCA-reduced OES features + discharge parameters.

- Principle Component Analysis is a dimensionality reduction technique that utilizes covariance matrix and solves its eigenvector as PCs.

$$C = \frac{1}{n-1} X^T X, \quad PCs = \{v, Cv = \lambda v\}$$

- Both linear and non-linear models are tested: Ridge, PLS, SVR, XGBoost, RF, MLP, CNN.
- Three configs of features/parameters are used as input data for models.

Config	Input Features	Dimensions
A	OES only (PCA-reduced)	11
B	Discharge parameters only	4
C	OES + discharge parameters	15 (P1) / 17 (P3)

Table 1: Configuration of Data Input Construction

Phase 2: Hyperparameter tuning of nonlinear models (RF, MLP, CNN) using Bayesian optimization.

- Optuna is a framework for hyperparameter tuning. It use Tree-structured Parzen Estimator to optimize hyperparameter for each model.
- RF, MLP and CNN are tuned with over 100 trails per model-config pair.

Phase 3: Manually construction input features based on plasma chemistry domain-knowledge.

- Construct physically meaningful features based on plamsa chemistry domain knowledge, instead of PCA-selected OES components.
- Update three training input configs with new features.

ID	Feature	Type	Species	Role
F1	I(309 nm)	Intensity	OH	Direct H_2O_2 precursor
F2	I(777 nm)	Intensity	O atom	CO_2 dissociation product
F3	I(656 nm)	Intensity	Hα	H_2O dissociation degree
F4	I(486 nm)	Intensity	Hβ	H_2O dissociation degree
F5	I(337 nm)	Intensity	N_2 SPS	High-energy e^- marker[2]
F6	I(406 nm)	Intensity	CO_2^+	CO_2 ionization degree
F7	I(516 nm)	Intensity	C_2	Possible Deep decomposition
F8	$\int$(306–312)	Band integral	OH 0-0 band	Robust OH emission
F9	$\int$(398–412)	Band integral	CO_2^+ FDB	Aggregate ionization
F10	$\int$(460–500)	Band integral	CO + Hβ	CO + H_2O dissociation
F11	I(309)/I(656)	Ratio	OH/Hα	Relative OH availability
F12	I(777)/I(309)	Ratio	O/OH	Relative OH availability
F13	I(656)/I(486)	Ratio	Hα/Hβ	Balmer decrement - e^- marker[3]

Table 2: Manually Selected OES Features

Phase 4: Interpretability analysis and feature reduction via multi-model consensus and backward elimination

- Evaluate feature importance for each model via different methods.

Model	Importance Method	What It Measures
Ridge	Absolute standardised regression coefficients	Linear contribution of each feature to the predicted target
PLS	VIP (Variable Importance in Projection)	Combined contribution across all PLS latent components, weighted by each component's explanatory power
RF	Permutation importance	Decrease in model accuracy when a single feature column is randomly shuffled
MLP	Mean |SHAP| value	Average marginal contribution of each feature across all possible feature coalitions

Table 3: Methods for Evaluating Feature Importance

- Statistically evaluate R^2 confidence interval of Ridge and PLS using bootstrap resampling.
- Analyze model's essential learning outcome using permutation test.
- Remove low representative features via eliminating one-by-one (backward elimination) or category-by-category (category ablation), to find the best performed model and config.

Results

Principle component analysis on 701 OES frequencies requires 11 components to reach 95% cumulative variance, and first 5 PCs capture ~74% of variance.

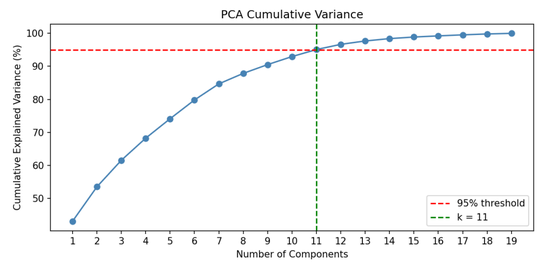

Figure 3: Principle Components Analysis Variance Curve

However, in phase 1, discharge parameter-only configurations dominate for most models, and OES-only inputs performs poorly.

After tuning parameters in phase 2, R^2 for all models are improved universally. MLP shows the most significant recovery from -1.13 to 0.33 for Config C and reaches 0.86 for Config B. CNN with config C remains the best OES-utilizing model.

The most pivotal breakthrough is in phase 3. Feature engineering dramatically improved Config C for MLP model from 0.33 to 0.80 and Ridge model from -0.17 to 0.80.

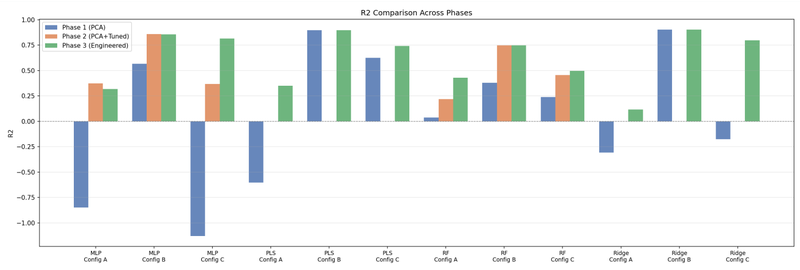

Figure 3: R^2 Performance for Each Phase

Phase 4 discovers that CO_2 flow rate and its OES band integral are two most important factor that affect H_2O_2 yield. Meanwhile, bootstrap confidence interval analysis shows that there is no statistic difference between Ridge config C and MLP config C.

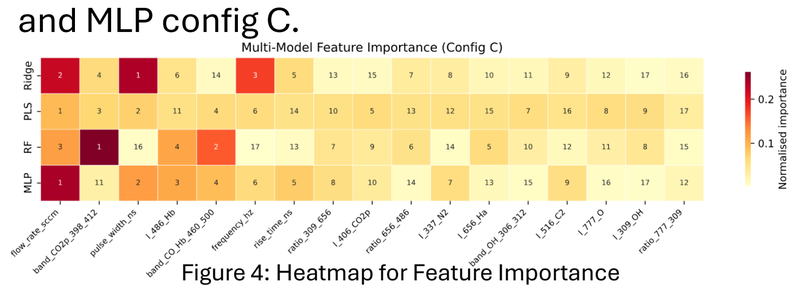

Figure 4: Heatmap for Feature Importance

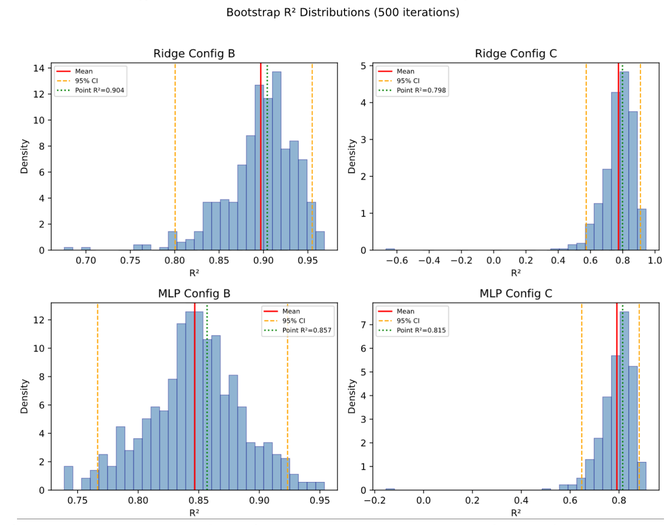

Figure 5: Bootstrap Confidence Interval Evaluation (Ridge vs MLP)

Permutation test validates that Ridge model successfully captures mapping relationship from X->Y with *p<0.00005*.

After eliminating features one by one reversely, the best model and input is found to be Ridge with 3 OES ratios + discharge parameter.

Rank	Model	Config	R^2	Features	Note
1	Ridge	C (3 OES ratio)	0.920	7	Category ablation
2	Ridge	C (1 OES ratio)	0.918	5	Backward elim. peak
3	Ridge	B	0.904	4	Phase 1 baseline
4	PLS	B	0.898	4	Phase 1 baseline
5	Ridge	C (4 OES)	0.899	8	Backward elim. n=4 peak
6	MLP	B	0.857	4	Phase 2 tuned
7	MLP	C	0.815	17	Phase 3 tuned
8	Ridge	C (full, Phase 3)	0.798	17	Phase 3 starting point
9	CNN	C	0.770	705	Phase 2 tuned

Table 4: Best Performed Models and Configs across Phases

Conclusion

This study demonstrates that domain-knowledge-driven feature engineering is the decisive factor in enabling accurate machine learning prediction of H_2O_2 yield from nanosecond pulsed CO_2 bubble plasma.

Replacing PCA with physically meaningful OES features dramatically improved model performance (Ridge Config C R^2 from −0.17 to 0.80).

Interpretability analysis identified CO_2 flow rate and its OES band integral as the most influential predictors. Permutation testing confirmed genuine predictive power ($p < 0.00005$), and a parsimonious Ridge model with just 3 OES ratios and discharge parameters matched complex neural networks, demonstrating that interpretable linear models suffice when features are well-engineered.

[1] Y. Gao, R. Zhou, L. Hong, B. Chen, J. Sun, R. Zhou, and Z. Liu, "Efficient synthesis of CO and H_2O_2 via nanosecond pulsed CO_2 bubble discharge," *J. Phys. D: Appl. Phys.*, vol. 57, no. 37, Sep. 2024, doi: 10.1088/1361-6463/ad5569.
[2] P. Paris, M. Aints, F. Valk, T. Plank, A. Haljaste, K. V. Kozlov, and H.-E. Wagner, "Intensity ratio of spectral bands of nitrogen as a measure of electric field strength in plasmas," J. Phys. D: Appl. Phys., vol. 38, no. 21, pp. 3894–3899, 2005, doi: 10.1088/0022-3727/38/21/010.
[3] H. R. Griem, Principles of Plasma Spectroscopy. Cambridge, U.K.: Cambridge University Press, 1997, ch. 7.